

Hannah Christina Arjonillo

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Data scientist with 6 years of industry experience in medtech and healthcare, spanning physiological time-series ML, behavioral text and topic modeling, and health outcomes modeling. Currently a Physics PhD student at UP Diliman with research interests in sociophysics and complex systems.

Experience

Data Scientist, BioIntellisense Inc.

Remote / Golden, CO | Dec 2024 – Apr 2026

- Led the exploration of a core body temperature estimation algorithm from skin and ambient temperature signals:
 - Supported clinical trial study design and data collection, informed by literature reviews and model iterations
 - Built a rule-based on/off-body detector using skin temperature rate-of-change thresholds to isolate on-body segments with equilibration transients
 - Fitted a physics-informed thermal model (constant, linear, and two exponential terms at different time scales) to extract features
 - Fed thermal features into a Ridge-CV regression with polynomial feature expansion and degree search
 - Conducted error analysis and validation against clinical reference measurements
- Benchmarked a vendor temperature model against the internal model using Bland-Altman and correlation analyses on clinical trial data; conducted error analyses on study limitations, informing algorithm selection and vendor discussions
- Contributed to clinical outcomes analyses on large-scale EMR data — built propensity-score-matching pipelines with calipered nearest-neighbor matching, Poisson log-link regressions for mortality and length-of-stay, subgroup interaction models, and balance diagnostics across hospital and hospital-to-home cohorts
- Developed candidate algorithms for wearable on/off-body detection and binary sleep/wake classification using multimodal signals (HR, HRV, skin temperature, accelerometry, ECG contact quality)
- Analyzed ECG-PPG pulse arrival time for cuffless blood pressure estimation, building subject-wise models and limits-of-agreement evaluations
- Constructed reusable ICD/DRG/MDC and Elixhauser comorbidity mappings from CMS reference tables for confounding control across analyses

Tools: Python, Databricks, MLflow, Claude Code

Machine Learning Scientist, Sequelae PH Inc. / Lore Health LLC

Remote / San Diego, CA | Dec 2022 – Nov 2024

- Designed and productionized topic models and other algorithms that aggregate human-generated content from the platform (Lore Health app / community.lorehealthcare.com) into featured central ideas called Lore
- Aided in the development of various human-in-the-loop processes, including automating tasks that transfer manual files to SQL Server tables, collaborating with software developers and writers, and working with Snorkel (software with human annotation and model development features)
- Built and iterated on models that describe how conversations evolve: for example, whether a section of text describes abstract or concrete ideas, or whether it shares a personal experience or asks for others' experiences
- Contributed to internal text processing and cloud utility packages

Tools: Python, Microsoft SQL Server, Google Cloud Platform, Airflow, Docker, REST API, Snorkel, Backstage, OpenAI API

Insighting Expert, Advanced Analytics Division, Enterprise Data Office, Globe Telecom, Inc.
Taguig City, Philippines | Jun 2022 – Dec 2022

- Developed an algorithm to spatially aggregate cell site activity and features based on their longitude and latitude
- Produced Tableau dashboards and presented spatial analysis findings to senior leadership

Tools: Python, Tableau

Lecturer (Part-time), National Institute of Physics, University of the Philippines - Diliman
Diliman, Quezon City, Philippines | Feb 2022 – May 2022

- Taught introductory physics and laboratory courses on electromagnetism to undergraduate students online

Data Scientist, OptumLabs, Inc. / UnitedHealth Group Research and Development (UHG R&D)

Cebu City, Philippines | Jan 2020 – Mar 2022

- Used NLP techniques to explore user reviews on a particular app acquired by UnitedHealth Group
- Developed a classification model that uses economic factors and health factors to predict whether a patient will soon develop Type 2 Diabetes, and compared its results to existing literature
- Contributed to the development of an epidemiological model (SIRD) to predict the cost of COVID-19 testing and the corresponding strategy to optimize the cost based on location infection rates, and developed the backend for a public-facing COVID-19 testing strategy simulator
- Investigated non-standard requests and problems using equation-based modeling, machine learning methods, statistics, and network science, implemented in Python and/or R
- Created reports, processed tables, and/or interactive dashboards to provide resources and tools to internal and external stakeholders

Tools: Google BigQuery, Python, R, Dash

Data Scientist Apprentice, UnitedHealth Group Research and Development (UHG R&D)
Makati, Philippines | Jul 2019 – Dec 2019

- BPI-UHG Data Science (BUDS) Training Program
- Acquired foundations in statistics, machine learning, and programming for data science research

Tools: Google BigQuery, Python, R, Tableau

Skills

- **Programming:** Python, R, Arduino C/C++
- **Data engineering:** Databricks, Google BigQuery, Microsoft SQL Server, Airflow
- **Model deployment:** FastAPI, Docker, Google Cloud Run
- **Dashboard deployment:** Streamlit, Dash (Python), CSS
- **AI-assisted workflows:** Claude Code, OpenAI API
- **Development:** GitHub / git, Jupyter Notebook, MLflow, Snorkel, PyCharm, RStudio, Backstage
- **Office:** LaTeX, Microsoft Office, Google Docs Suite

Publications

- Euste, J. L., Arjonillo, H. C. and Saloma, C. (2024) Time-resolved culture maps derived from the Integrated Values Survey data (1981-2022). **Physica A: Statistical Mechanics and its Applications**

659:130317, <https://doi.org/10.1016/j.physa.2024.130317>

- Arjonillo, H. C. and Saloma, C. (2023) Monte Carlo analysis of focused Gaussian beam in scattering media: Curvature correction and Mie scattering. **Optics Communications** **535:129314**, <https://doi.org/10.1016/j.optcom.2023.129314>. International Publication Award (IPA) 2023 recipient, University of the Philippines - Diliman

Education

- **University of the Philippines - Diliman**, PhD Physics (August 2025 - Present). Research focus: Hyperspectral unmixing applied to senatorial voting patterns in the Philippines. Adviser: May T. Lim, PhD
- **University of the Philippines - Diliman**, MS Physics (January 2021 - June 2024). Thesis title: Measuring intra-country cultural diversity. Adviser: Caesar A. Saloma, PhD
- **University of the Philippines - Diliman**, BS Applied Physics (Instrumentation Physics), Cum Laude (August 2014 - June 2019). Thesis title: Towards a more accurate Monte Carlo analysis of propagating focused Gaussian beam in scattering media. Adviser: Caesar A. Saloma, PhD

Music Education

- **Associated Board of the Royal Schools of Music (ABRSM)**, Diploma in Music Performance (Violin) (2016)
- **Associated Board of the Royal Schools of Music (ABRSM)**, Grade 8: Violin (Distinction), Piano (Merit) (2014)
- **Associated Board of the Royal Schools of Music (ABRSM)**, Grade 5: Theory of Music (Merit) (2012)

Conferences & Workshops

- **42nd Samahang Pisika ng Pilipinas Physics Conference** (June 2024, Online) — Contributor and presenter. H. C. Arjonillo, J. L. Euste, and C. Saloma. Measuring cultural diversity within countries with ellipses, contour maps, and distances, Proceedings of the Samahang Pisika ng Pilipinas 42, SPP-2024-1D-01 (2024). <https://proceedings.spp-online.org/article/view/SPP-2024-1D-01>
- **Workshop on Quantitative Human Ecology, Abdus Salam International Centre for Theoretical Physics** (July 2022, Online) — Participant. Workshop bringing together scientists from economics, ecology, demography, physics, complex systems science, climatology, epidemiology, social sciences, and intelligent systems to promote wider integration of methods across disciplines tackling sustainability challenges.
- **39th Samahang Pisika ng Pilipinas Physics Conference** (October 2021, Online) — Contributor and presenter. H. C. Arjonillo, J. L. Euste, and C. Saloma. Correlating cultural values with COVID-19 outcomes, Proceedings of the Samahang Pisika ng Pilipinas 39, SPP-2021-PB-20 (2021). <https://paperview.spp-online.org/proceedings/article/view/SPP-2021-PB-20>
- **37th Samahang Pisika ng Pilipinas Physics Conference** (June 2019, Bohol, Philippines) — Contributor and presenter. H. C. Arjonillo and C. Saloma. Curvature correction and Mie scattering in Monte Carlo analysis of propagating focused Gaussian beam in scattering media, Proceedings of the Samahang Pisika ng Pilipinas 37, SPP-2019-PB-20 (2019). <https://paperview.spp-online.org/proceedings/article/view/SPP-2019-PB-20>
- **36th Samahang Pisika ng Pilipinas Physics Conference** (June 2018, Palawan, Philippines) — Contributor and presenter. H. C. Arjonillo and C. Saloma. Preserving photon detection and average power in random scattering media and curvature correction in vectorized particle Monte Carlo simulation of focused Gaussian beam, Proceedings of the Samahang Pisika ng Pilipinas 36, SPP-2018-2C-06 (2018). <https://paperview.spp-online.org/proceedings/article/view/SPP-2018-2C-06>